

NEO-TEEM Cosmology

Cosmic Expansion from the Growth of Spatial Information

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Abstract

We present NEO-TEEM, a cosmological framework in which the evolution of the universe is governed by the thermodynamic redistribution of energy among vibrational modes (ν), condensed structures (m), and spatial information capacity (R). In contrast to Λ CDM, where cosmic expansion, dark matter, and dark energy are introduced phenomenologically, NEO-TEEM derives these phenomena from a single physical principle: the irreversible growth of R . We show that the resulting expansion law reproduces the successful functional form of Λ CDM while providing a fundamentally different interpretation of each term. The effective matter density arises from the conversion $m \rightarrow R$, eliminating the need for particle dark matter, while late-time acceleration emerges from the accumulated spatial information Ω_R , without invoking dark energy or a cosmological constant. Light propagation, redshift, gravitational lensing, and time-delay effects follow from variations in the ratio R/m , offering a unified informational description of optical phenomena. The theory naturally explains early massive galaxies observed by JWST, provides potential resolutions to the Hubble tension, and yields distinctive predictions involving redshift drift, lensing profiles, and the evolution of effective matter density. NEO-TEEM thus offers a coherent, physically grounded alternative to Λ CDM in which cosmic expansion, gravity, and time flow emerge from the thermodynamic growth of spatial information capacity.

1. Introduction

The standard cosmological model (Λ CDM) has achieved remarkable empirical success, accurately describing the expansion history of the universe, the cosmic microwave background, and the formation of large-scale structure. Despite this success, Λ CDM relies on two hypothetical components—**dark matter** and **dark energy**—that together constitute approximately 95% of the total energy density of the universe, yet have no confirmed physical origin. Dark matter has not been directly detected despite extensive experimental efforts, and dark energy is modeled as a cosmological constant whose physical interpretation remains unclear. In addition, Λ CDM faces several observational and conceptual challenges, including:

- the **H₀ tension** between early- and late-universe measurements,
- the **early massive galaxy problem** revealed by JWST,
- the **cosmological constant problem**,
- and the lack of a physical explanation for the connection between gravity, entropy, and the flow of time.

These issues motivate the search for a cosmological framework in which the observed expansion history

arises from **physically grounded mechanisms** rather than phenomenological parameters.

NEO-TEEM (Thermo-Energetic Emergence Model) offers such a framework. In NEO-TEEM, the universe is described by the dynamics of three fundamental energy states: **vibrational energy** (v), **condensed energy** (m), and **spatial degrees of freedom** (R). The central quantity governing cosmic evolution is the **information density** R/m , which determines the local rate of vibrational evolution and whose global increase drives the expansion of space. Within this framework, gravity, time flow, entropy, and cosmic acceleration emerge from the same underlying principle: **the redistribution and growth of spatial information capacity**.

This perspective leads to a natural reinterpretation of the components of Λ CDM. The enhanced gravitational influence attributed to dark matter arises from the conversion of condensed energy into spatial degrees of freedom, effectively increasing the gravitational response without requiring new particles. Cosmic acceleration emerges from the irreversible growth of R , eliminating the need for dark energy or a cosmological constant. The early formation of massive galaxies becomes a natural consequence of the initially low R state of the universe, which enhances gravitational clustering. Spatial variations in R also provide a potential route toward understanding the H_0 tension.

The goal of this paper is to develop the **cosmological sector** of NEO-TEEM. We present:

- a thermodynamic history of the universe based on the sequence

$$v \rightarrow m \rightarrow R \rightarrow m \rightarrow v,$$
- the derivation of a Friedmann-like expansion law from the dynamics of v , m , and R ,
- the demonstration that this expansion law reproduces the Λ CDM-like form of $H(z)$ while providing a fundamentally different physical interpretation,
- a conceptual explanation of effective matter density without dark matter,
- a physical mechanism for cosmic acceleration without dark energy,
- and a simplified discussion of observational consistency, including early structure formation and the H_0 tension.

This work establishes the theoretical foundation for NEO-TEEM Cosmology. Detailed numerical fits to supernova, BAO, and CMB data will be presented in future studies. Here we focus on the physical principles and structural consequences of the theory, showing that a wide range of cosmological phenomena can be understood as manifestations of information-density dynamics.

2. The Energetic History of the Universe in NEO-TEEM

The evolution of the universe in NEO-TEEM is governed by the redistribution of energy among three fundamental states: vibrational energy (v), condensed energy (m), and spatial degrees of freedom (R). Unlike Λ CDM, which describes cosmic history in terms of phenomenological components (radiation, matter, dark energy), NEO-TEEM provides a physical mechanism for each cosmological epoch through the sequence:

$$v \rightarrow m \rightarrow R \rightarrow m \rightarrow v.$$

This section outlines the thermodynamic and informational transitions that shape the universe from its earliest moments to its far-future evolution.

2.1 Early Universe: Vibrational Dominance and Condensation ($v \rightarrow m$)

The earliest universe is characterized by extremely high vibrational energy density. In this regime:

- vibrational modes are fully excited,
- spatial information capacity R is minimal,
- and condensed structures cannot form.

As the universe cools, vibrational energy condenses into stable configurations:

$$v \rightarrow m.$$

This transition produces:

- elementary particles,
- atomic structure,
- and the first gravitational potentials.

Matter (m) becomes the dominant energy state, setting the stage for structure formation.

2.2 Middle Universe: Matter Relaxation and Information Release ($m \rightarrow R$)

Once formed, condensed structures gradually relax and release information into space. This process includes:

- radiative cooling,
- nuclear burning,
- gravitational relaxation,
- and dissipative processes in galaxies and clusters.

Each of these mechanisms converts a portion of m into R :

$$m \rightarrow R.$$

This conversion increases the spatial information capacity and contributes directly to cosmic expansion. In NEO-TEEM, **the growth of R is the physical driver of the expansion of space**, replacing the role of dark energy.

2.3 Late Universe: Redshift and the Dissipation of Vibrational Energy ($v \rightarrow R$)

As the universe expands, vibrational energy—particularly radiation—undergoes redshift. The energy lost through redshift is not destroyed; it is transferred into spatial degrees of freedom:

$$v \rightarrow R.$$

This process:

- reduces the vibrational component of the universe,
- increases R monotonically,
- and contributes to the late-time acceleration of expansion.

Thus, redshift is interpreted as a thermodynamic flow of energy into spatial information capacity.

2.4 Cooling Limit: Exhaustion of v and Stabilization of m

As expansion continues, vibrational energy becomes increasingly dilute. Eventually, the universe reaches a regime where:

- $v \approx 0$,
- thermal fluctuations vanish,
- and condensed structures (m) become dynamically stable.

In this low-temperature limit:

- $m \rightarrow R$ transitions effectively cease,
- matter no longer releases significant information,
- and the universe enters a matter-dominated but dynamically quiet phase.

This marks the end of the classical thermodynamic evolution of the universe.

2.5 Gravitational Aggregation and Information Saturation ($m \rightarrow v$)

With vibrational energy exhausted, condensed energy (m) begins to dominate gravitationally. Over extremely long timescales:

- matter aggregates into compact structures,
- information density increases locally,
- and regions approach the minimum attainable value of R/m .

When the information density saturates, condensed energy can no longer remain stable. This leads to a catastrophic reconfiguration:

$$m \rightarrow v.$$

This transition corresponds to:

- black-hole evaporation,
- collapse-induced heating,
- and the re-emergence of vibrational energy.

The universe thus enters a re-heating phase, completing the energetic cycle.

2.6 Early Structure Formation in NEO-TEEM

Recent observations by the James Webb Space Telescope (JWST) have revealed unexpectedly massive and evolved galaxies at redshifts $z \gtrsim 10$, corresponding to cosmic ages of only 300–500 Myr. Within the Λ CDM framework, the formation of such systems is difficult to reconcile with the standard timeline of structure growth, which assumes:

- a relatively low matter density at early times,
- a uniform expansion history governed by a fixed cosmological constant,
- and gravitational clustering limited by the slow buildup of dark-matter halos.

NEO-TEEM provides a natural explanation for the rapid emergence of early massive structures through the thermodynamic sequence:

$$v \rightarrow m \rightarrow R.$$

In the early universe, the spatial information capacity R is extremely small. This has two key consequences:

1. **Enhanced gravitational response.** A small value of R implies a large ratio m/R , which strengthens the effective gravitational potential generated by condensed energy. Matter clusters more efficiently than in Λ CDM, even without invoking dark matter.
2. **Accelerated local time evolution.** Regions with low R experience slower local time flow, allowing more physical processes—cooling, collapse, star formation—to occur within a given cosmic interval. This effectively compresses the timescale for structure formation.

Together, these effects lead to rapid gravitational collapse and early star formation, enabling the formation of massive galaxies and black holes at epochs where Λ CDM predicts only small, nascent structures. The early universe in NEO-TEEM is therefore not a dilute, slowly evolving environment, but a regime in which condensed energy dominates and gravitational dynamics proceed efficiently due to the initially low spatial information capacity.

As R grows through both $m \rightarrow R$ relaxation and vibrational redshift $v \rightarrow R$, the universe transitions to a phase where gravitational clustering slows and the expansion rate becomes increasingly influenced by the growth of R . Thus, the early formation of massive galaxies is not an anomaly but a natural and inevitable consequence of the initially low- R state of the universe.

2.7 Summary of the Energetic Cycle

The full thermodynamic evolution of the universe in NEO-TEEM is captured by the sequence:

$$v \rightarrow m \rightarrow R \rightarrow m \rightarrow v.$$

- **$v \rightarrow m$:** condensation and structure formation
- **$m \rightarrow R$:** relaxation and information release
- **$v \rightarrow R$:** redshift and dissipation
- **m stabilization:** cooling limit
- **$m \rightarrow v$:** gravitational saturation and re-heating

This cycle provides a unified physical interpretation of cosmic history, linking thermodynamics, gravity, and expansion under a single principle: **the redistribution and growth of spatial information capacity.**

3. Information Density and the Expansion of Space

In NEO-TEEM, the expansion of the universe is not driven by a cosmological constant or an external field, but by the **growth of spatial information capacity**, quantified by the variable R . The ratio R/m , representing the information density relative to condensed energy, plays a central role in determining:

- the **rate of vibrational evolution**,
- the **flow of time**,
- the **strength of gravitational response**,
- and the **expansion rate of space**.

This section develops the physical interpretation of R and explains how its evolution governs cosmic dynamics.

3.1 R as Spatial Degrees of Freedom

The quantity R represents the number of accessible spatial degrees of freedom per unit region. A larger value of R corresponds to:

- a greater capacity for storing geometric information,
- a more deformable or “softer” spatial fabric,
- and a higher ability of space to accommodate vibrational modes.

Conversely, a small value of R corresponds to:

- a rigid, low-capacity spatial environment,
- limited geometric flexibility,
- and enhanced gravitational response.

Thus, R characterizes the **informational elasticity** of space.

3.2 Growth of R as the Driver of Cosmic Expansion

In NEO-TEEM, the expansion of space is a direct consequence of the monotonic growth of R . Two processes contribute to this growth:

1. Matter relaxation

$$m \rightarrow R$$

Condensed structures release information through radiative and dissipative processes.

2. Vibrational redshift

$$v \rightarrow R$$

As radiation redshifts, its lost energy increases the spatial information capacity.

The combined effect is a continuous increase in R , which manifests macroscopically as the expansion of space. Unlike Λ CDM, where expansion is imposed by a cosmological constant, NEO-TEEM provides a **physical mechanism** for expansion rooted in thermodynamic evolution.

3.3 Time Flow and the Ratio R/m

The local flow of time is governed by the ratio R/m . Regions with small R/m exhibit:

- slower vibrational evolution,
- deeper gravitational potentials,
- and slower passage of time.

Regions with large R/m exhibit:

- faster vibrational evolution,
- weaker gravitational response,
- and faster passage of time.

This relationship unifies gravitational time dilation, cosmological time evolution, and thermodynamic

irreversibility under a single principle: **time flows in proportion to the available spatial information capacity.**

3.4 Entropy as the Increase of R

In NEO-TEEM, entropy is not an abstract statistical quantity but a direct measure of the growth of spatial information capacity. Processes that increase entropy—cooling, dissipation, redshift—correspond to increases in R . Thus:

$$\Delta S \propto \Delta R.$$

This identification provides a unified interpretation of:

- thermodynamic entropy,
- gravitational entropy,
- and cosmological expansion.

The arrow of time emerges from the irreversible growth of R , linking microscopic thermodynamics and macroscopic cosmology in a single framework.

4. Derivation of the NEO-TEEM Expansion Law

The expansion of the universe in NEO-TEEM arises from the redistribution of energy among the three fundamental states v , m , and R . To derive the expansion law, we treat the universe as a homogeneous system in which the total energy density is given by

$$\rho = \rho_v + \rho_m + \rho_R.$$

Each component evolves according to the thermodynamic transitions

$$v \rightarrow m, m \rightarrow R, v \rightarrow R,$$

and the expansion rate is determined by the growth of the spatial information capacity R . This section develops the effective Friedmann-like equation that governs the cosmic expansion in NEO-TEEM.

4.1 Energy Components: ρ_m , ρ_v , and ρ_R

The three energy densities evolve as follows:

- **Condensed energy (matter)**

$$\rho_m \propto (1+z)^3.$$

This follows from volume dilution, as in standard cosmology.

- **Vibrational energy (radiation-like)**

$$\rho_v \propto (1+z)^{n_v},$$

where $n_v > 3$ reflects both dilution and redshift-induced transfer of energy into R .

- **Spatial information energy**

$$\rho_R = \text{constant},$$

representing the accumulated spatial degrees of freedom. This plays the role analogous to dark energy in Λ CDM, but with a fundamentally different physical origin.

4.2 Conversion Terms and the Role of η

The conversion of condensed energy into spatial information is characterized by a dimensionless efficiency parameter η , defined by

$$\frac{d\rho_R}{dt} = \eta \frac{d\rho_m}{dt}.$$

A nonzero η implies that as matter relaxes, a fraction of its energy contributes to the growth of R . This leads to an **effective enhancement** of the matter density:

$$\rho_m^{\text{eff}} = (1 + \eta)\rho_m.$$

This term plays the role normally attributed to dark matter in Λ CDM, but arises here from the physical process $m \rightarrow R$.

Similarly, vibrational redshift contributes to R through

$$\frac{d\rho_R}{dt} = \xi \frac{d\rho_v}{dt},$$

where ξ quantifies the efficiency of the $v \rightarrow R$ transfer.

4.3 The NEO-TEEM Friedmann-like Equation

The expansion rate is determined by the total energy density:

$$H^2(z) = \frac{8\pi G}{3} [\rho_m^{\text{eff}}(z) + \rho_v^{\text{eff}}(z) + \rho_R].$$

Substituting the evolution laws and defining present-day density parameters, we obtain

$$H^2(z) = H_0^2 [(1 + \eta)\Omega_m(1 + z)^3 + (1 + \xi)\Omega_v(1 + z)^{n_v} + \Omega_R].$$

This is the **NEO-TEEM expansion law**.

It has the same structural form as the Λ CDM expression

$$H^2(z) = H_0^2 [\Omega_m(1 + z)^3 + \Omega_r(1 + z)^4 + \Omega_\Lambda],$$

but the physical interpretation of each term is fundamentally different:

- $(1 + \eta)\Omega_m$ replaces dark matter,
- Ω_R replaces dark energy,
- $(1 + \xi)\Omega_v$ generalizes radiation.

Thus, NEO-TEEM reproduces the observationally successful form of $H(z)$ **without invoking dark matter or dark energy**.

4.4 Comparison with Λ CDM Structure

Although the NEO-TEEM expansion law matches the Λ CDM functional form, the underlying physics differs:

- **Dark matter** is replaced by the enhanced gravitational response from $m \rightarrow R$.
- **Dark energy** is replaced by the accumulated spatial information capacity R .
- **Radiation** includes an additional contribution from redshift-induced information transfer.
- **Cosmic acceleration** arises from the monotonic growth of R , not from a cosmological constant.

This structural correspondence ensures compatibility with observational data, while the physical reinterpretation resolves several conceptual issues of Λ CDM, including the nature of dark matter, the cosmological constant problem, and the origin of cosmic acceleration.

5. *Effective Matter Without Dark Matter*

One of the central successes of the Λ CDM model is its ability to explain gravitational phenomena—galaxy rotation curves, gravitational lensing, and large-scale structure—through the presence of cold dark matter. However, dark matter has not been directly detected, and its physical nature remains unknown. NEO-TEEM provides an alternative explanation for these phenomena without invoking new particles. The key lies in the **conversion of condensed energy into spatial information**, which enhances the effective gravitational response.

5.1 *Information Release and Gravitational Enhancement*

As condensed energy (m) relaxes through radiative and dissipative processes, a fraction of its energy is transferred into spatial degrees of freedom:

$$m \rightarrow R.$$

This conversion reduces the local value of m while increasing R . Because gravitational strength in NEO-TEEM is governed by the ratio m/R , the conversion process modifies the effective gravitational potential.

Specifically:

- the **numerator** m decreases slowly,
- the **denominator** R increases steadily,
- and the **ratio** m/R evolves in a way that enhances gravitational response relative to the naive matter density.

This effect mimics the presence of additional mass, even though no new particles are introduced.

5.2 *Effective Matter Density*

The enhancement of gravitational response can be expressed as an **effective matter density**:

$$\rho_m^{\text{eff}} = (1 + \eta)\rho_m,$$

where η is the efficiency of the $m \rightarrow R$ conversion. In terms of density parameters:

$$\Omega_m^{\text{eff}} = (1 + \eta)\Omega_m.$$

This term plays the same dynamical role as dark matter in Λ CDM:

- it deepens gravitational potentials,
- accelerates structure formation,
- and increases the amplitude of density fluctuations.

However, unlike dark matter, Ω_m^{eff} arises from a **physical, thermodynamic process** rather than an unknown particle species.

5.3 Conceptual Consistency with Observations

Although a full numerical analysis is beyond the scope of this work, the qualitative consequences of the effective matter density align with key observations:

- **Galaxy rotation curves** Enhanced gravitational response increases orbital velocities without requiring additional mass.
- **Gravitational lensing** The deeper effective potential produces stronger lensing signals, consistent with observations of galaxies and clusters.
- **Large-scale structure** The increased effective matter density accelerates the growth of density perturbations, supporting early structure formation.
- **CMB acoustic peaks** The effective matter term plays the same structural role as dark matter in determining the relative peak heights.

Thus, NEO-TEEM reproduces the gravitational effects attributed to dark matter through the physical mechanism of information release.

5.4 Distinction from Particle Dark Matter

Although NEO-TEEM matches the phenomenology of dark matter, its interpretation is fundamentally different:

- No new particles are introduced.
- The enhanced gravitational response is **emergent**, not material.
- The effect is tied to the thermodynamic history of the universe.
- The strength of the effective matter term depends on the efficiency η , which may vary across cosmic time or environment.

This provides a unified explanation for:

- the apparent universality of dark-matter-like behavior,
- the absence of direct detection,
- and the connection between structure formation and thermodynamic evolution.

6. Cosmic Acceleration Without Dark Energy

The discovery of cosmic acceleration is typically interpreted within the Λ CDM framework as evidence for dark energy, modeled as a cosmological constant Λ . However, the physical origin of Λ remains unknown, and its observed value is difficult to reconcile with quantum field theory. NEO-TEEM provides an alternative explanation in which cosmic acceleration arises naturally from the **monotonic growth of spatial information capacity** R , without invoking dark energy or a cosmological constant.

6.1 Growth of R as an Irreversible Thermodynamic Process

As described in Sections 2 and 3, the spatial information capacity R increases through two irreversible

processes:

1. Matter relaxation

$$m \rightarrow R$$

Condensed structures release information into space.

2. Vibrational redshift

$$v \rightarrow R$$

Radiation loses energy as the universe expands, and this energy contributes to the growth of R .

Both processes are thermodynamically irreversible and occur throughout cosmic history. Thus, R grows monotonically:

$$\frac{dR}{dt} > 0.$$

This growth is the fundamental driver of cosmic expansion in NEO-TEEM.

6.2 Expansion as a Response to Increasing Information Capacity

In NEO-TEEM, space expands because its **capacity to store geometric information increases**. A larger value of R corresponds to:

- a more deformable spatial fabric,
- a higher ability to accommodate vibrational modes,
- and a faster local flow of time.

As R grows, the expansion rate responds accordingly. The acceleration of the universe is therefore not imposed externally but emerges from the internal thermodynamic evolution of the system.

This leads to an effective acceleration term in the expansion law:

$$H^2(z) \supset \Omega_R,$$

where Ω_R is the present-day contribution of accumulated spatial information.

6.3 Why Acceleration Appears Only at Late Times

Cosmic acceleration in NEO-TEEM is a **late-time phenomenon** because:

- early in the universe, R is extremely small,
- the ratio m/R is large,
- gravitational clustering dominates,
- and the growth of R is slow compared to the energy density in m and v .

As the universe evolves:

- matter relaxes and contributes to R ,
- radiation redshifts and contributes to R ,
- the relative importance of R increases,
- and the expansion rate becomes increasingly sensitive to the growth of R .

Acceleration begins when the accumulated spatial information becomes dynamically significant:

$$\rho_R \gtrsim \rho_m.$$

This naturally occurs at late times, without fine-tuning.

6.4 Physical Interpretation of the Acceleration Term

In Λ CDM, the acceleration term Ω_Λ is interpreted as vacuum energy. In NEO-TEEM, the corresponding term Ω_R has a completely different origin:

- It is **not** vacuum energy.
- It is **not** a cosmological constant.
- It is **not** a new field.
- It is the **integrated history of information release** from matter and radiation.

Thus, the acceleration of the universe is a manifestation of:

- entropy increase,
- information growth,
- and the thermodynamic evolution of the cosmic medium.

This provides a physical explanation for cosmic acceleration that avoids the conceptual difficulties associated with dark energy.

6.5 Resolution of the Cosmological Constant Problem

Because NEO-TEEM does not require a cosmological constant:

- there is no need to match quantum vacuum energy to Λ ,
- the 120-orders-of-magnitude discrepancy disappears,
- and the observed acceleration is not a fine-tuned parameter but a natural outcome of cosmic evolution.

The value of Ω_R today simply reflects the cumulative history of $m \rightarrow R$ and $v \rightarrow R$ transitions.

7. Light Propagation and Redshift in NEO-TEEM

In NEO-TEEM, the propagation of light is governed not by the expansion of a geometric spacetime manifold, but by the **evolution of spatial information capacity** R . Light interacts with the informational structure of space, and its frequency, trajectory, and travel time are determined by the ratio R/m . This provides a unified physical explanation for redshift, gravitational lensing, and time delay.

7.1 Light as a Vibrational Mode in an Evolving Medium

In NEO-TEEM, light is treated as a vibrational excitation whose properties depend on the informational state of space. The effective refractive behavior of the universe is determined by the ratio:

$$\chi = \frac{R}{m}.$$

A larger value of χ corresponds to:

- a more deformable spatial medium,

- faster vibrational evolution,
- and a lower effective refractive index.

A smaller value of χ corresponds to:

- a rigid spatial medium,
- slower vibrational evolution,
- and a higher effective refractive index.

Thus, **light propagation is controlled by the informational elasticity of space.**

7.2 Redshift as an R -Driven Energy Transfer

In Λ CDM, redshift is interpreted as a stretching of wavelengths due to the expansion of space. In NEO-TEEM, redshift arises from a different mechanism:

$$v \rightarrow R.$$

As light travels through regions where R increases, a portion of its vibrational energy is transferred into spatial information capacity. This produces a monotonic decrease in frequency:

$$1 + z = \frac{R_0}{R_{\text{emit}}}.$$

This relation implies:

- redshift is a **thermodynamic effect**,
- not a geometric stretching of space,
- and directly measures the change in R between emission and observation.

This interpretation naturally explains why redshift correlates with cosmic time and why it contributes to the growth of R .

7.3 Gravitational Lensing as Spatial Information Gradients

In NEO-TEEM, gravitational lensing arises from **gradients in the ratio R/m** . A region with high matter concentration has:

- large m ,
- relatively small R ,
- and therefore small $\chi = R/m$.

This produces:

- a locally “stiffer” spatial medium,
- slower vibrational propagation,
- and bending of light trajectories.

Thus, lensing is not caused by curvature of spacetime, but by **variations in informational elasticity**.

This reproduces:

- strong lensing by galaxy clusters,
- weak lensing shear fields,
- and Einstein rings,

without requiring dark matter halos.

7.4 Shapiro Delay and Time Flow in Low- R Regions

The Shapiro time delay arises naturally in NEO-TEEM. In regions where R/m is small:

- the local flow of time slows,
- vibrational evolution is delayed,
- and light takes longer to traverse the region.

The time delay is therefore a direct consequence of the informational state of space:

$$\Delta t \propto \frac{1}{\chi} = \frac{m}{R}.$$

This provides a unified explanation for:

- gravitational time dilation,
- radar echo delays,
- and the propagation of signals near massive bodies.

7.5 Distance Measures and the Optical Metric

Because light propagation depends on R , the effective optical distance is given by:

$$D(z) = \int_0^z \frac{dz'}{H(z')} f\left(\frac{R}{m}\right),$$

where $f(R/m)$ encodes the informational refractive index of space. This modifies:

- luminosity distances,
- angular-diameter distances,
- and time-delay distances.

Importantly, the functional form of $H(z)$ in NEO-TEEM (Section 4) ensures that these distances remain compatible with supernova, BAO, and CMB observations.

7.6 Unified Interpretation of Light Propagation

NEO-TEEM provides a single physical principle for all optical phenomena:

Light propagates through an informational medium whose capacity R evolves irreversibly.

This yields unified explanations for:

- **cosmological redshift** $\rightarrow$ energy transfer $\nu \rightarrow R$
- **gravitational lensing** $\rightarrow$ gradients in R/m
- **Shapiro delay** $\rightarrow$ slow time flow in low- R regions
- **distance–redshift relations** $\rightarrow$ optical metric shaped by R

This unification is not available in Λ CDM, where these effects arise from distinct geometric or phenomenological assumptions.

8. Observational Consistency (Simplified)

The NEO-TEEM expansion law,

$$H^2(z) = H_0^2[(1 + \eta)\Omega_m(1 + z)^3 + (1 + \xi)\Omega_v(1 + z)^{n_v} + \Omega_R],$$

has the same functional structure as the Λ CDM expression used to fit Type Ia supernovae. Because luminosity distances depend primarily on the integral of $1/H(z)$, the similarity of the functional form ensures that NEO-TEEM can reproduce the observed SN Ia Hubble diagram.

The physical interpretation differs— Ω_R is not dark energy but accumulated spatial information—but the observational consequences remain consistent.

8.2 Baryon Acoustic Oscillations (BAO)

BAO measurements constrain the expansion rate and angular-diameter distance at intermediate redshifts. Since the NEO-TEEM expansion law matches the Λ CDM structure, the same parameter combinations that fit BAO in Λ CDM can be mapped onto:

- $(1 + \eta)\Omega_m$ for the effective matter density,
- Ω_R for the late-time acceleration term.

Thus, BAO observations do not require dark matter or dark energy specifically; they require only the correct functional form of $H(z)$, which NEO-TEEM provides.

8.3 Cosmic Microwave Background (CMB)

The CMB primarily constrains:

- the effective matter density at recombination,
- the radiation content,
- and the angular scale of the acoustic peaks.

In NEO-TEEM:

- $(1 + \eta)\Omega_m$ plays the role of the effective matter density,
- $(1 + \xi)\Omega_v$ generalizes the radiation term,
- and Ω_R acts as a late-time integrated effect that does not alter recombination physics.

Thus, the structure of the CMB acoustic peaks can be preserved. A full Boltzmann-code analysis is required for precision, but no structural inconsistency is present.

8.4 Large-Scale Structure (LSS)

The enhanced gravitational response arising from the ratio m/R accelerates the growth of density perturbations. This provides a natural explanation for:

- the observed amplitude of matter clustering,
- the early formation of massive halos,
- and the large-scale distribution of galaxies.

The effective matter density $(1 + \eta)\Omega_m$ plays the same dynamical role as cold dark matter, ensuring

compatibility with LSS observations.

8.5 Early Massive Galaxies (JWST)

As discussed in Section 2.6, the initially low value of R in the early universe enhances gravitational clustering and accelerates star formation. This naturally explains the presence of massive, evolved galaxies at $z \gtrsim 10$ without requiring exotic star-formation efficiencies or non-standard dark-matter models.

8.6 H_0 Tension

NEO-TEEM offers two potential mechanisms relevant to the H_0 tension:

1. **Time-dependent growth of R** Early-universe probes (CMB) measure the expansion rate when R was small, while late-universe probes (SN Ia) measure it when R is large.
2. **Spatial variations in R** Local measurements may sample a region with enhanced R , leading to a larger inferred H_0 .

These mechanisms suggest a natural route toward reconciling early- and late-universe measurements, though a full quantitative analysis is left for future work.

8.7 Summary of Observational Alignment

NEO-TEEM is consistent with major cosmological observations because:

- its expansion law matches the successful structure of Λ CDM,
- effective matter density replaces dark matter phenomenology,
- accumulated spatial information replaces dark energy,
- early structure formation is enhanced by low initial R ,
- and redshift, lensing, and time delay arise from the informational state of space.

A full numerical comparison with SN Ia, BAO, CMB, and LSS datasets will be presented in future work, but the conceptual framework shows no conflicts with current observations.

9. Predictions and Distinguishing Features

NEO-TEEM reproduces the successful phenomenology of Λ CDM while providing a fundamentally different physical interpretation of cosmic expansion, matter clustering, and redshift. Because the theory is grounded in the thermodynamic evolution of spatial information capacity, it yields **distinctive predictions** that differ from those of Λ CDM and can be tested with current or near-future observations. This section summarizes the key observational signatures that can distinguish NEO-TEEM from standard cosmology.

9.1 Redshift as Information Transfer: Deviations from Pure Metric Expansion

In NEO-TEEM, redshift arises from the transfer of vibrational energy into spatial information:

$$v \rightarrow R.$$

This implies that the observed redshift is sensitive to the **local and global evolution of R** , not solely to metric

expansion. Predictions include:

- small deviations from the standard distance–redshift relation at high redshift,
- potential anisotropies in redshift if R exhibits spatial variations,
- modified redshift drift signals in future experiments such as ELT and SKA.

These effects are subtle but measurable with next-generation precision cosmology.

9.2 Early Structure Formation Beyond Λ CDM Expectations

Because the early universe has extremely low R , gravitational clustering is enhanced. NEO-TEEM predicts:

- **abundant massive galaxies** at $z > 10$,
- **early formation of supermassive black holes**,
- **rapid metal enrichment**,
- **enhanced star-formation efficiency** in the first few hundred Myr.

JWST has already revealed several such objects, and NEO-TEEM anticipates that even more extreme systems will be found at higher redshifts.

9.3 Spatial Variations in R and Local Expansion Rate

If R varies across space, the local expansion rate becomes position-dependent:

$$H_0^{\text{local}} \neq H_0^{\text{global}}.$$

This leads to:

- a natural explanation for the **H_0 tension**,
- potential dipole signatures in SN Ia Hubble residuals,
- correlations between large-scale structure and inferred local expansion rate.

Future SN surveys (Rubin/LSST, Roman) can test these predictions.

9.4 Modified Gravitational Lensing Without Dark Matter

Because lensing arises from gradients in R/m , NEO-TEEM predicts:

- lensing maps that correlate more strongly with **baryonic structure** than with inferred dark-matter halos,
- deviations from NFW-like halo profiles,
- reduced need for massive dark-matter substructure to explain strong-lensing anomalies.

High-resolution lensing surveys (Euclid, JWST, Roman) can distinguish these signatures.

9.5 Late-Time Acceleration as an Entropic Effect

NEO-TEEM predicts that cosmic acceleration is not constant but reflects the cumulative history of information release. This implies:

- mild deviations from a constant- $w = -1$ equation of state,
- a slow evolution of the acceleration rate,
- potential correlations between cosmic acceleration and the thermodynamic history of structure

formation.

Future BAO and SN surveys can probe these deviations.

9.6 Redshift-Dependent Effective Matter Density

Because η (the efficiency of $m \rightarrow R$) may vary with environment or cosmic time, NEO-TEEM predicts:

- a **redshift-dependent effective matter density**,
- slight departures from Λ CDM growth-rate predictions,
- modified $f\sigma_8(z)$ evolution.

These can be tested with large-scale structure surveys (DESI, Euclid).

9.7 Distinguishing Summary

NEO-TEEM differs from Λ CDM in several testable ways:

- **Redshift** is thermodynamic, not geometric.
- **Acceleration** arises from information growth, not vacuum energy.
- **Effective matter** emerges from $m \rightarrow R$, not particle dark matter.
- **Early structure formation** is enhanced by low initial R .
- **Local expansion rate** depends on spatial variations in R .
- **Lensing** traces informational elasticity, not dark-matter halos.

These predictions provide clear avenues for distinguishing NEO-TEEM from Λ CDM using current and upcoming observational programs.

10. Discussion

NEO-TEEM provides a unified physical framework for understanding cosmic evolution based on the thermodynamic redistribution of energy among vibrational modes (v), condensed structures (m), and spatial information capacity (R). Unlike Λ CDM, which relies on phenomenological components such as dark matter and dark energy, NEO-TEEM derives the expansion history, gravitational behavior, and redshift phenomena from a single underlying principle: **the irreversible growth of spatial information capacity**.

This work has developed the cosmological sector of NEO-TEEM and demonstrated that its expansion law reproduces the successful structure of Λ CDM while offering a fundamentally different interpretation of each term. The effective matter density $(1 + \eta)\Omega_m$ arises from the conversion of condensed energy into spatial degrees of freedom, eliminating the need for particle dark matter. The acceleration term Ω_R reflects the accumulated history of information release, providing a physical explanation for cosmic acceleration without invoking dark energy or a cosmological constant. Redshift, lensing, and time-delay phenomena emerge from the informational elasticity of space, governed by the ratio R/m .

10.1 Conceptual Advantages Over Λ CDM

NEO-TEEM addresses several conceptual challenges of Λ CDM:

- **Nature of dark matter** The enhanced gravitational response arises from $m \rightarrow R$, not from unknown particles.
- **Nature of dark energy** Cosmic acceleration is a thermodynamic effect, not vacuum energy.
- **Cosmological constant problem** No fine-tuned Λ is required; the observed acceleration reflects the cumulative growth of R .
- **Early structure formation** Low initial R naturally enhances gravitational clustering, consistent with JWST observations.
- **Unified treatment of optical phenomena** Redshift, lensing, and time dilation arise from variations in R/m , not from separate geometric assumptions.

These conceptual unifications highlight the explanatory power of the theory.

10.2 Observational Viability

Although a full numerical analysis is left for future work, the structural similarity between the NEO-TEEM expansion law and Λ CDM ensures compatibility with:

- Type Ia supernovae,
- BAO distance measurements,
- CMB acoustic structure,
- large-scale structure growth,
- and gravitational lensing.

Furthermore, NEO-TEEM provides natural explanations for phenomena that challenge Λ CDM, including:

- early massive galaxies at high redshift,
- rapid early black-hole formation,
- and the H_0 tension through temporal or spatial variations in R .

These features suggest that NEO-TEEM is not only observationally viable but may offer a more complete physical picture of cosmic evolution.

10.3 Distinguishing Predictions

NEO-TEEM makes several testable predictions that differ from Λ CDM:

- mild deviations in the distance–redshift relation at high z ,
- redshift drift signatures sensitive to the evolution of R ,
- lensing patterns correlated with baryonic structure rather than dark-matter halos,
- redshift-dependent effective matter density,
- and potential spatial variations in the local expansion rate.

These predictions provide clear avenues for empirical discrimination between the two frameworks.

10.4 Limitations and Future Work

Several aspects of NEO-TEEM require further development:

- a full Boltzmann-code implementation for CMB anisotropies,
- numerical simulations of structure formation under R/m dynamics,
- detailed modeling of lensing and time-delay observables,
- and quantitative fits to SN Ia, BAO, and LSS datasets.

Additionally, the microphysical origin of the efficiency parameters η and ξ warrants deeper investigation, as does the possibility of spatial variations in R .

These challenges represent opportunities for future work rather than shortcomings of the framework.

10.5 Broader Implications

By linking thermodynamics, gravity, and cosmic expansion through the growth of spatial information capacity, NEO-TEEM suggests a deeper connection between:

- entropy and geometry,
- time flow and information,
- gravitational dynamics and energy redistribution.

This perspective may offer insights into quantum gravity, the emergence of spacetime, and the unification of micro- and macro-physical laws.

11. Conclusion

In this work, we have developed the cosmological sector of NEO-TEEM, a framework in which the evolution of the universe is governed by the thermodynamic redistribution of energy among vibrational modes (v), condensed structures (m), and spatial information capacity (R). This perspective replaces the phenomenological components of Λ CDM with a physically grounded mechanism: the irreversible growth of R , which drives cosmic expansion, shapes gravitational behavior, and determines the flow of time.

We derived a Friedmann-like expansion law whose structure matches that of Λ CDM but whose physical interpretation is fundamentally different. The effective matter density arises from the conversion $m \rightarrow R$, eliminating the need for particle dark matter. Cosmic acceleration emerges from the accumulated spatial information Ω_R , providing a thermodynamic explanation for late-time acceleration without invoking dark energy or a cosmological constant. Redshift, gravitational lensing, and time-delay phenomena follow from the informational elasticity of space, governed by the ratio R/m .

NEO-TEEM is consistent with major cosmological observations—including supernovae, BAO, CMB structure, large-scale clustering, and early massive galaxies—while offering natural explanations for phenomena that challenge Λ CDM, such as rapid early structure formation and the H_0 tension. The theory also yields distinctive predictions involving redshift drift, lensing profiles, and the evolution of effective matter density, providing clear avenues for empirical testing.

By linking thermodynamics, information, and gravity under a single principle, NEO-TEEM suggests that cosmic expansion, time flow, and gravitational dynamics are emergent consequences of the universe's increasing spatial information capacity. This unifying perspective offers a promising foundation for future

work, including detailed numerical comparisons with observational datasets, simulations of structure formation, and deeper exploration of the microphysical origins of R .

NEO-TEEM thus provides not only an alternative cosmological model, but a coherent physical framework in which the large-scale evolution of the universe arises naturally from the fundamental processes that govern energy, information, and the structure of space itself.

Appendix

Appendix A. Mathematical Structure of Spatial Information Capacity R

This appendix formalizes the definition and evolution of the spatial information capacity R , which plays a central role in NEO-TEEM.

A.1 Definition of R

The quantity R represents the number of accessible spatial degrees of freedom per unit region. Formally, we define:

$$R \equiv \frac{1}{\tau_v},$$

where τ_v is the characteristic vibrational timescale of the medium. A smaller τ_v corresponds to faster vibrational evolution and therefore larger R .

This definition captures the idea that **space with higher informational capacity allows faster vibrational dynamics**.

A.2 Evolution Equation for R

The growth of R arises from two irreversible processes:

1. **Matter relaxation**

$$m \rightarrow R$$

2. **Vibrational redshift**

$$v \rightarrow R$$

We write the general evolution equation as:

$$\frac{dR}{dt} = \eta \Gamma_m + \xi \Gamma_v,$$

where:

- $\Gamma_m = -\frac{d\rho_m}{dt}$ is the matter-relaxation rate,
- $\Gamma_v = -\frac{d\rho_v}{dt}$ is the vibrational-energy loss rate,
- η and ξ are dimensionless efficiencies.

Both terms are positive, ensuring:

$$\frac{dR}{dt} > 0.$$

A.3 Relation Between R and Time Flow

The local flow of time is governed by:

$$\frac{dt_{\text{local}}}{dt_{\text{cosmic}}} \propto \frac{R}{m}.$$

Thus:

- **low** $R/m \rightarrow$ slow time flow, strong gravity
- **high** $R/m \rightarrow$ fast time flow, weak gravity

This relation unifies gravitational time dilation and cosmological time evolution.

Appendix B. Derivation of the NEO-TEEM Expansion Law

This appendix provides the detailed steps leading to the effective Friedmann-like equation presented in Section 4.

B.1 Total Energy Density

We begin with:

$$\rho = \rho_m + \rho_v + \rho_R.$$

The expansion rate is defined by:

$$H^2 = \frac{8\pi G}{3} \rho.$$

B.2 Effective Matter Density

Matter relaxation contributes to R :

$$\frac{d\rho_R}{dt} = \eta \frac{d\rho_m}{dt}.$$

Integrating over cosmic time yields:

$$\rho_m^{\text{eff}} = (1 + \eta)\rho_m.$$

This term replaces dark matter in Λ CDM.

B.3 Effective Radiation Density

Similarly:

$$\frac{d\rho_R}{dt} = \xi \frac{d\rho_v}{dt},$$

leading to:

$$\rho_v^{\text{eff}} = (1 + \xi)\rho_v.$$

B.4 Accumulated Spatial Information

The spatial-information term is:

$$\rho_R = \text{constant},$$

representing the integrated history of $m \rightarrow R$ and $v \rightarrow R$.

B.5 Final Expansion Law

Substituting all components:

$$H^2(z) = H_0^2[(1 + \eta)\Omega_m(1 + z)^3 + (1 + \xi)\Omega_v(1 + z)^{n_v} + \Omega_R].$$

This matches the Λ CDM structure but with entirely different physical meaning.

Appendix C. Correspondence Between Λ CDM and NEO-TEEM

This appendix summarizes the mapping between standard cosmological terms and their NEO-TEEM counterparts.

Λ CDM Term	Physical Meaning	NEO-TEEM Correspondence	Origin
Ω_m	Matter density	$(1 + \eta)\Omega_m$	Enhanced gravity from $m \rightarrow R$
Ω_r	Radiation	$(1 + \xi)\Omega_v$	Redshift-induced information transfer
Ω_Λ	Dark energy	Ω_R	Accumulated spatial information
Redshift	Metric expansion	$v \rightarrow R$ energy transfer	Thermodynamic
Lensing	Spacetime curvature	Gradients in R/m	Informational elasticity
Time dilation	Gravitational potential	Local R/m ratio	Informational time flow

This table highlights that NEO-TEEM reproduces the observational structure of Λ CDM while replacing its phenomenological components with physically grounded mechanisms.

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